TRIBHUVAN UNIVERSITY

# Padmakanya Multiple Campus

Bagbazar, Kathmandu

# LAB REPORT ON NETWORK AND SYSTEM ADMINISTRATION (BIT451)


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# Contents

**1. Install Linux system in the virtual environment. All the following task should be performed in this Linux system. ..... 1**

**2. Configure following Network Setting (as additional): ..... 6**

**3. Schedule a job to print "Hello world!" in every 15 minutes daily. ..... 9**

**4. Make a directory "pk_bit" under the root (/) and this directory should have following properties: ..... 10**

**5. Add a user in the system with following properties: ..... 12**

**6. Configure a DHCP server for the network 172.32.255.0/24 with following options: ..... 13**

**7. Configure a DNS server for the domain "example.com" and the server should be following hosts pointing to the following IP addresses. Any email for this domain should be delivered to mail.example.com. ..... 15**

**8. Configure a Web Server to host host1.example.com and host2.example.com. The host1.example.com should serve from the folder /var/www/host1 and host2.example.com should serve from the folder /var/www/host2. ..... 17**

**9. Configure a Mail Server to receive and deliver the mail for the domain "example.com". The mail server should reject any email which has the content "SPAM" in the body of the email. ..... 20**

**1. Install Linux system in the virtual environment. All the following task should be performed in this Linux system.**

- **The system should have utilize LVM feature.**
- **The system should have a super user with the password of your Roll Number.**

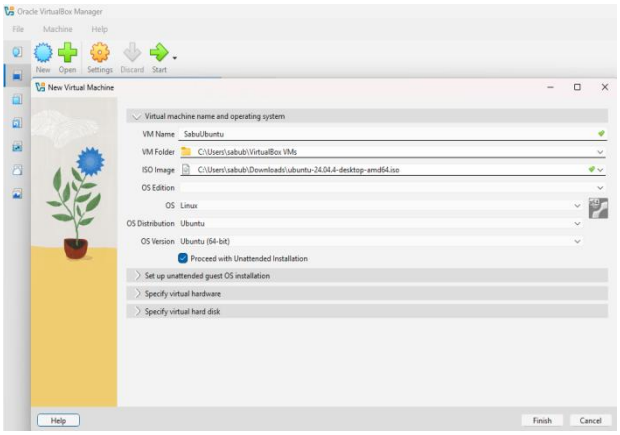
To install a Linux system in a virtual environment with LVM support and configure a superuser with a specified password.

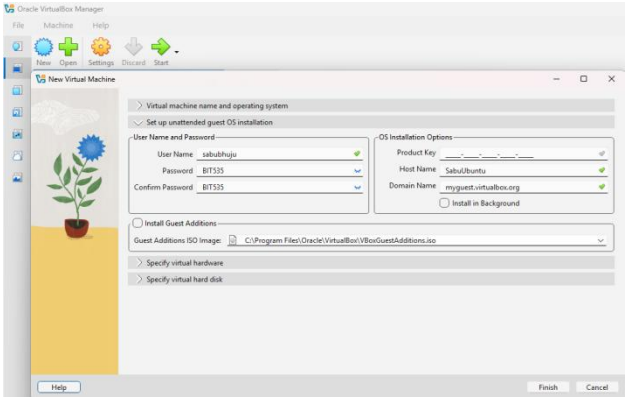
**a. Install Virtualization Software**

- First, install Oracle VM VirtualBox on the host computer to create a virtual environment for installing the Ubuntu Linux operating system.

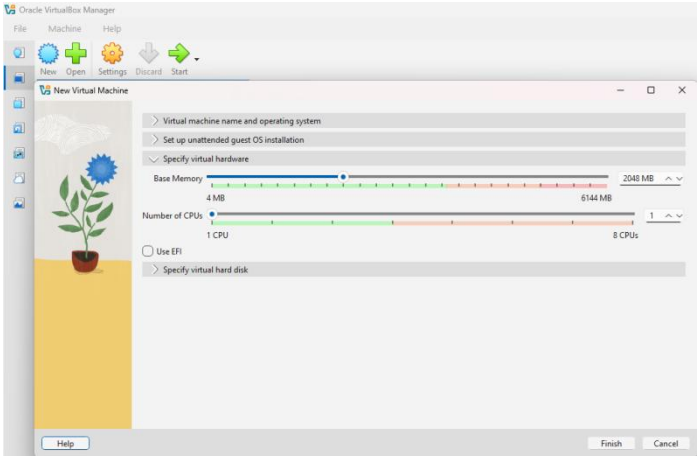
**b. Create a New Virtual Machine**

- Open VirtualBox.
- Click New to create a new virtual machine.
- Provide the name of the virtual machine.
- Select the operating system type as Linux and version as Ubuntu.




- Allocate RAM and CPU sources for the virtual machine.

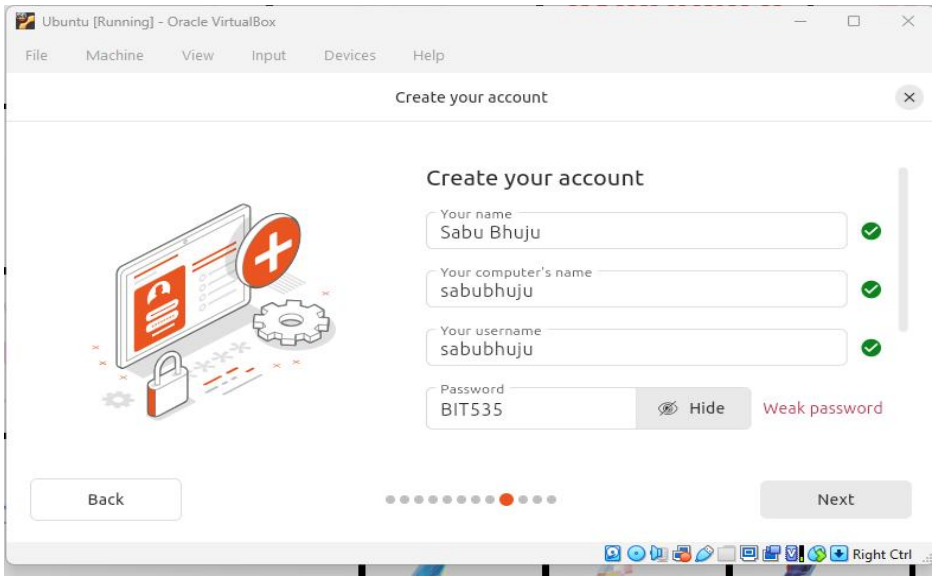


**c. Insatll Ubuntu Opearting System.**

- Downlaod the Ubuntu ISO file.
- Attach the ISO file to the virtual machine as a bootable disk.
- Start the virtual machine to begin the installation process.
- Select language, keyboard layout and time zone.
- Complete the installation process.

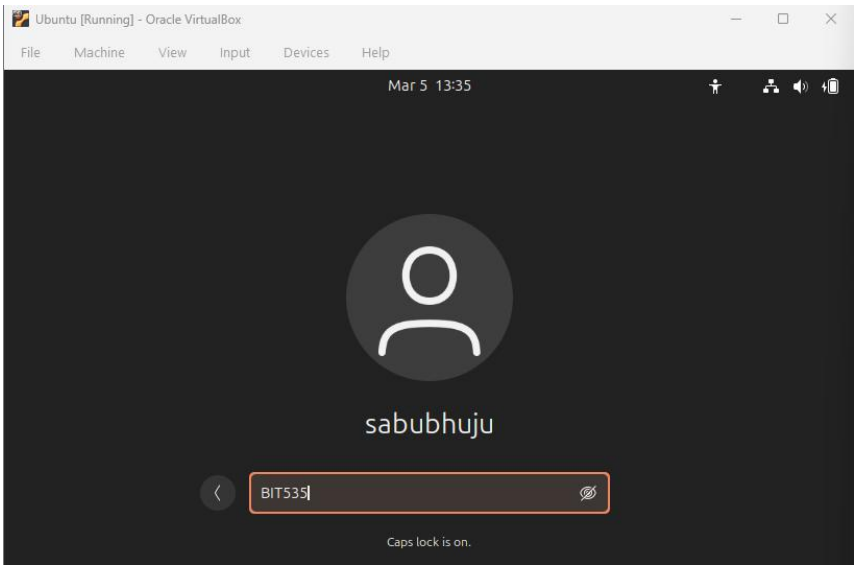
**d. Create the User Account (Superuser)**

- A user account is created with required password as following:




**e. Login to Ubuntu System**

After the installation was completed, the system was restarted and the user logged into the Ubuntu system.



## f. Configure LVM Using Terminal

After logging into the system, the Logical Volume Manager (LVM) was configured using terminal commands.

- Check your disk

```
sabubhuju@sabubhuju:~$ lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0    7:0    0     4K  1 loop /snap/bare/5
loop1    7:1    0    74M  1 loop /snap/core22/2292
loop2    7:2    0 251.7M  1 loop /snap/firefox/7766
loop3    7:3    0 251.6M  1 loop /snap/firefox/7836
loop4    7:4    0  18.5M  1 loop /snap/firmware-updater/210
loop5    7:5    0 531.4M  1 loop /snap/gnome-42-2204/247
loop6    7:6    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop7    7:7    0  10.8M  1 loop /snap/snap-store/1270
loop8    7:8    0  48.1M  1 loop /snap/snapd/25935
loop9    7:9    0   576K  1 loop /snap/snapd-desktop-integration/343
sda      8:0    0    25G  0 disk
├─sda1   8:1    0     1M  0 part
└─sda2   8:2    0    25G  0 part /
sdb      8:16   0    25G  0 disk
sr0     11:0    1  50.9M  0 rom  /media/sabubhuju/VBox_GAs_7.2.6
```

- Created Physical Volume (PV) from available disk space.

```
sabubhuju@sabubhuju:~$ sudo pvcreate /dev/sdb
  Physical volume "/dev/sdb" successfully created.
```

- Created Volume Group (VG) from the physical volume.

```
sabubhuju@sabubhuju:~$ sudo vgcreate sabu_vg /dev/sdb
  Volume group "sabu_vg" successfully created
```

- Created Logical Volume (LV) from the volume group.

```
sabubhuju@sabubhuju:~$ sudo lvcreate -L 5G -n sabu_lv sabu_vg
  Logical volume "sabu_lv" created.
```

- Formatted and mounted the logical volume for system use.

```
sabubhuju@sabubhuju:~$ sudo mkfs.ext4 /dev/sabu_vg/sabu_lv
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 1ca461bb-b059-4df7-8139-f09b08d5306b
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

sabubhuju@sabubhuju:~$ sudo mkdir /lvm_data
sabubhuju@sabubhuju:~$ sudo mount /dev/sabu_vg/sabu_lv /lvm_data
```

- Verify LVM:

```
sabubhuju@sabubhuju:~$ sudo pvdisplay
  --- Physical volume ---
  PV Name               /dev/sdb
  VG Name               sabu_vg
  PV Size               25.00 GiB / not usable 4.00 MiB
  Allocatable           yes
  PE Size               4.00 MiB
  Total PE              6399
  Free PE               5119
  Allocated PE          1280
  PV UUID               DZAP18-xsjt-LgoW-CW6X-taYG-6mN3-O9O5gK

sabubhuju@sabubhuju:~$ sudo vgdisplay
  --- Volume group ---
  VG Name               sabu_vg
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  2
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                1
  Open LV               1
```

```
sabubhuju@sabubhuju:~$ sudo lvdisplay
  --- Logical volume ---
  LV Path                /dev/sabu_vg/sabu_lv
  LV Name                sabu_lv
  VG Name                sabu_vg
  LV UUID                9RvTUw-Wi3P-Sya5-pfrr-uYJ4-VJEO-ErfXyQ
  LV Write Access        read/write
  LV Creation host, time sabubhuju, 2026-02-28 20:03:17 +0545
  LV Status              available
  # open                 1
  LV Size                5.00 GiB
  Current LE             1280
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     16384
  Block device           252:0

sabubhuju@sabubhuju:~$ df -h
Filesystem                   Size  Used Avail Use% Mounted on
tmpfs                        348M  1.5M  346M   1% /run
/dev/sda2                     25G  9.2G   15G  40% /
tmpfs                        1.7G     0  1.7G   0% /dev/shm
tmpfs                        5.0M  8.0K  5.0M   1% /run/lock
tmpfs                        348M  116K  347M   1% /run/user/1000
/dev/sr0                      51M   51M     0 100% /media/sabubhuju/VBox_GAs_7.2.6
/dev/mapper/sabu_vg-sabu_lv  4.9G   24K  4.6G   1% /lvm_data
sabubhuju@sabubhuju:~$
```

**2. Configure following Network Setting (as additional):**

- **IP address should be the member of 172.16.0.0/16 family.**
- **Last number of IP address should be the last two digit of your roll number (e.g. if your roll number is BIT215 then the last number of IP should be 15).**
- **Gateway should be 172.16.254.254.**
- **DNS should be 8.8.8.8 and 8.8.4.4**

To configure a static IP address in the 172.16.0.0/16 network range with the correct gateway and DNS settings, and verify network connectivity.

**a. Check or identify the network interface**

```
sabubhuju@sabubhuju:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group
 default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel stat
e UP group default qlen 1000
    link/ether 08:00:27:2c:51:54 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.108/24 brd 192.168.0.255 scope global dynamic noprefix
route enp0s3
       valid_lft 7169sec preferred_lft 7169sec
    inet6 fe80::a00:27ff:fe2c:5154/64 scope link
       valid_lft forever preferred_lft forever
```

**b. Configure the IP address(My roll number is BIT535; so using lst two digits 35).**

```
sabubhuju@sabubhuju: ~
sabubhuju@sabubhuju:~$ sudo ip addr add 172.16.1.35/16 dev enp0s3
sabubhuju@sabubhuju:~$ sudo ip route add default via 172.16.254.254
```

**c. Configure DNS by editing /etc/resolve.conf.**

```
sabubhuju@sabubhuju:~$ sudo nano /etc/resolv.conf

sabubhuju@sabubhuju:~$ sudo cat ^C
sabubhuju@sabubhuju:~$ sudo cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-res
olved(8).
# Do not edit.

nameserver 127.0.0.53
options edns0 trust-ad
search .

nameserver 8.8.8.8
nameserver 8.8.4.4
```

**d. Verify the configuration.**

```
sabubhuju@sabubhuju:~$ sudo ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group
default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state
 UP group default qlen 1000
    link/ether 08:00:27:2c:51:54 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute en
p0s3
       valid_lft 85817sec preferred_lft 85817sec
    inet 172.16.1.35/16 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fd17:625c:f037:2:c6cc:8bd4:17f7:e942/64 scope global temporary d
ynamic
       valid_lft 86355sec preferred_lft 14355sec
    inet6 fd17:625c:f037:2:a00:27ff:fe2c:5154/64 scope global dynamic mngt
mpaddr
```

**e. Ping and verify:**

```
sabubhuju@sabubhuju:~$ ping 8.8.8.8

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=36.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=38.4 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=115 time=36.9 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=115 time=37.9 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=115 time=39.3 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=115 time=39.5 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=115 time=35.6 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=115 time=50.2 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=115 time=42.3 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=115 time=38.8 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=115 time=33.9 ms
64 bytes from 8.8.8.8: icmp_seq=67 ttl=115 time=39.7 ms
64 bytes from 8.8.8.8: icmp_seq=68 ttl=115 time=45.3 ms
64 bytes from 8.8.8.8: icmp_seq=69 ttl=115 time=35.4 ms
64 bytes from 8.8.8.8: icmp_seq=70 ttl=115 time=37.1 ms
^C
--- 8.8.8.8 ping statistics ---
70 packets transmitted, 68 received, 2.85714% packet loss, time 69173ms
rtt min/avg/max/mdev = 33.651/65.411/758.662/95.170 ms
sabubhuju@sabubhuju:~$
```

## 3. Schedule a job to print “Hello world!” in every 15 minutes daily.

To schedule a recurring job that prints “Hello world!” every 15 minutes using the cron scheduler.

**a. Open the terminal**

**b. Edit the user’s crontab file.**

```
sabubhuju@sabubhuju: ~
sabubhuju@sabubhuju:~$ crontab -e
no crontab for sabubhuju - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.tiny
  3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
```

**c. Add the following line to schedule the job every 15 minutes.**

```
Mar 5 14:08
sabubhuju@sabubhuju: ~
  GNU nano 7.2          /tmp/crontab.SWxyKq/crontab *
*/15 * * * * echo "Hello World!" >> /home/sabubhuju/hello.txt
```

**d. Save and exit**

**e. Verify the cron jobs**

```
sabubhuju@sabubhuju:~$ crontab -l
*/15 * * * * echo "Hello World!" >> /home/sabubhuju/hello.txt
```

**f. Wait for 15 minutes or monitor the file in real time using tail -f /home/sabubhuju/hello.txt**

```
sabubhuju@sabubhuju:~$ ls
Desktop    Downloads  Music     Public  Templates
Documents  hello.txt  Pictures  snap    Videos
sabubhuju@sabubhuju:~$ sudo tail -f /home/sabubhuju/hello.txt
Hello World!
Hello World!
Hello World!
```

**4. Make a directory "pk_bit" under the root (/) and this directory should have following properties:**

- **This directory should not be accessible to all users except the member of 'pk_bit' group.**
- **The member of 'pk_bit' group shall create a file inside this directory and should be readable/writeable to the all member of the 'pk_bit' group.**
- **Any new file/folders under this directory should have default group ownership of 'pk_bit' group.**
- **Any files under this directory can only be deleted by the owner user only.**

To create a directory /pk_bit with restricted access for all users except members of the pk_bit group, set default group ownership, and configure file permissions according to the requirements.

**a.Create a group named pk_bit**

```
sabubhuju@sabubhuju: ~
sabubhuju@sabubhuju:~$ sudo groupadd pk_bit
sabubhuju@sabubhuju:~$ getent group pk_bit
pk_bit:x:1004:
```

**b. Create a directory under root.**

```
sabubhuju@sabubhuju:~$ sudo mkdir /pk_bit
```

**c. Change the group ownership of the directory.**

```
sabubhuju@sabubhuju:~$ sudo chgrp pk_bit /pk_bit
sabubhuju@sabubhuju:~$
```

**d. Set directory permissions so only group members can access it:**

```
sabubhuju@sabubhuju:~$ sudo chmod 770 /pk_bit
sabubhuju@sabubhuju:~$
```

**f. Set the setgid bit so new files/folders inherit the group ownership pk_bit:**

```
sabubhuju@sabubhuju:~$ sudo chmod g+s /pk_bit
sabubhuju@sabubhuju:~$
```

**g. Add users to the pk_bit group**

```
sabubhuju@sabubhuju:~$ sudo usermod -aG pk_bit sabubhuju
sabubhuju@sabubhuju:~$
```

**h. Create a file inside the directory and set file permissions so all group members can read/write.**

```
sabubhuju@sabubhuju:~$ touch /pk_bit/testfile.txt
sabubhuju@sabubhuju:~$ chmod 660 /pk_bit/testfile.txt
sabubhuju@sabubhuju:~$
```

**i. Ensure only the owner can delete files inside /pk_bit**

```
sabubhuju@sabubhuju:~$ sudo chmod +t /pk_bit
sabubhuju@sabubhuju:~$
```

**j. Verify the permissions and ownership of the directory.**

```
sabubhuju@sabubhuju:~$ ls -ld /pk_bit
drwxrws--T 2 root pk_bit 4096 Mar  2 13:13 /pk_bit
sabubhuju@sabubhuju:~$
```

## 5. Add a user in the system with following properties:

- **User name should be your full name without any spaces.**
- **Password for the username should be same as username.**
- **The user should be a member of the "pk_bit" group.**

To create a new user in the Linux system with a specified username, password, and group membership.

**a. Verify the group exists or not. If not create the group.**

```
sabubhuju@sabubhuju: ~
sabubhuju@sabubhuju:~$ sudo groupadd pk_bit
sabubhuju@sabubhuju:~$ getent group pk_bit
pk_bit:x:1004:
```

**b. Create a new user and add to the pk_bit group**

```
sabubhuju@sabubhuju:~$ sudo useradd -m BHUJUSABU
```

**c. Set the user's password (same as the username)**

```
sabubhuju@sabubhuju:~$ echo "BHUJUSABU:BHUJUSABU" | sudo chpasswd
BAD PASSWORD: The password contains the user name in some form
```

**d. Add newly create user to the group.**

```
sabubhuju@sabubhuju:~$ sudo usermod -aG pk_bit BHUJUSABU
```

**e. Verify the user is created and belongs to the correct group**

```
sabubhuju@sabubhuju:~$ id BHUJUSABU
uid=1003(BHUJUSABU) gid=1005(BHUJUSABU) groups=1005(BHUJUSABU),1004(pk_bit)
sabubhuju@sabubhuju:~$
```

**6. Configure a DHCP server for the network 172.32.255.0/24 with following options:**

- **DHCP Client should get an unique IP within the range of 172.32.255.101 to 172.32.255.120 only**
- **The DHCP client should have Gateway of IP 172.32.255.254.**
- **The DHCP client should have DNS IP as 1.1.1.1**

To configure a DHCP server for the network 172.32.255.0/24 so that clients receive unique IPs within a specified range, with a defined gateway and DNS server.

**a. Install the DHCP server package**

```
sabubhuju@sabubhuju: ~
sabubhuju@sabubhuju:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://np.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://np.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,4
sabubhuju@sabubhuju:~$ sudo apt install isc-dhcp-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
isc-dhcp-server is already the newest version (4.4.3-P1-4ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 19 not upgraded.
```

b. **Edit the DHCP server configuration file and add the network configuration**

```
sabubhuju@sabubhuju:~$ sudo nano /etc/dhcp/dhcpd.conf
sabubhuju@sabubhuju:~$ sudo cat /etc/dhcp/dhcpd.conf

subnet 172.32.255.0 netmask 255.255.255.0 {
 range 172.32.255.101 172.32.255.120;
 option routers 172.32.255.254;
 option domain-name-servers 1.1.1.1;
 option subnet-mask 255.255.255.0;
 default-lease-time 600;
 max-lease-time 7200;
}
```

**c. Specify the interface the DHCP server should listen on in /etc/default/isc-dhcp-server**

```
sabubhuju@sabubhuju:~$ sudo nano /etc/default/isc-dhcp-server
sabubhuju@sabubhuju:~$ sudo cat /etc/default/isc-dhcp-server
INTERFACESv4="enp0s3"
INTERFACESv6=""
```

**d. Start, enable and verify the status of the DHCP server.**

```
sabubhuju@sabubhuju:~$ sudo systemctl restart isc-dhcp-server

sabubhuju@sabubhuju:~$ sudo systemctl enable isc-dhcp-server
Synchronizing state of isc-dhcp-server.service with SysV service script with /usr/lib/systemd/
systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable isc-dhcp-server
sabubhuju@sabubhuju:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled
)
     Active: active (running) since Sat 2026-02-28 20:41:20 +0545; 11s ago
       Docs: man:dhcpd(8)
   Main PID: 5155 (dhcpd)
      Tasks: 1 (limit: 4063)
     Memory: 3.7M (peak: 4.3M)
        CPU: 14ms
     CGroup: /system.slice/isc-dhcp-server.service
             └─5155 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /e
v
```

**7. Configure a DNS server for the domain "example.com" and the server should be following hosts pointing to the following IP addresses. Any email for this domain should be delivered to mail.example.com.**
**ns.example.com 172.32.255.1**
**mail.example.com 172.32.255.2**
**www.example.com 172.32.255.3**
**host1.example.com 172.32.255.4**
**host2.example.com 172.32.255.4**

To configure a DNS server for the domain example.com with multiple host entries and mail server configuration.

**a. Install BIND9**

```
sabubhuju@sabubhuju:~$ sudo apt install bind9 bind9utils bind9-doc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bind9-utils
Suggested packages:
  bind-doc
The following NEW packages will be installed:
  bind9 bind9-doc bind9-utils bind9utils
0 upgraded, 4 newly installed, 0 to remove and 13 not upgraded.
```

**b. Configure the BIND9 Zone**

```
sabubhuju@sabubhuju:~$ sudo nano /etc/bind/named.conf.local
sabubhuju@sabubhuju:~$ sudo cat /etc/bind/named.conf.local
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "example.com" IN{
      type master;
      file "/etc/bind/zones/db.example.com";
      allow-update { none; };
};
```

### c. Create the Zone File

```
sabubhuju@sabubhuju:~$ sudo nano /etc/bind/zones/db.example.com
sabubhuju@sabubhuju:~$ sudo cat /etc/bind/zones/db.example.com
$TTL    86400
@       IN      SOA   ns.example.com. admin.example.com. (
                      2026022801 ; Serial
                      3600       ; Refresh
                      1800       ; Retry
                      604800     ; Expire
                      86400 )    ; Minimum TTL
; Name servers
@     IN      NS    ns.example.com.

; Hosts
ns    IN      A     172.32.255.1
mail  IN      A     172.32.255.2
www   IN      A     172.32.255.3
host1 IN      A     172.32.255.4
host2 IN      A     172.32.255.4

; MX record
@     IN      MX 10  mail.example.com.
sabubhuju@sabubhuju:~$ sudo named-checkzone example.com /etc/bind/zones/db.example.com
zone example.com/IN: loaded serial 2026022801
OK
sabubhuju@sabubhuju:~$ sudo systemctl restart bind9
```

### d. Check Configuration and Restart BIND9

```
sabubhuju@sabubhuju:~$ sudo named-checkzone example.com /etc/bind/zones/db.example.com
zone example.com/IN: loaded serial 2026022801
OK
```

### e. Restart BIND9

```
sabubhuju@sabubhuju:~$ sudo systemctl restart bind9
```

### f. Test DNS Resolution

```
sabubhuju@sabubhuju:~$ dig @localhost example.com MX

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @localhost example.com MX
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 30120
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 34d9d692cc8da6540100000069a30db512e293766c08fb53 (good)
;; QUESTION SECTION:
;example.com.                    IN      MX

;; ANSWER SECTION:
example.com.            86400   IN      MX      10 mail.example.com.

;; ADDITIONAL SECTION:
mail.example.com.       86400   IN      A       172.32.255.2

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(localhost) (UDP)
;; WHEN: Sat Feb 28 21:30:57 +0545 2026
;; MSG SIZE  rcvd: 105
```

## 8. Configure a Web Server to host host1.example.com and host2.example.com. The host1.example.com should serve from the folder /var/www/host1 and host2.example.com should serve from the folder /var/www/host2.

To configure an Apache web server to host multiple virtual hosts (host1.example.com and host2.example.com) with separate document roots.

### a. Install Apache2 and verify status

```
sabubhuju@sabubhuju:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  libaprutil1t64
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64
0 upgraded, 8 newly installed, 0 to remove and 13 not upgraded.
sabubhuju@sabubhuju:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-02-28 21:58:37 +0545; 55s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 9156 (apache2)
      Tasks: 55 (limit: 4063)
     Memory: 5.5M (peak: 5.9M)
        CPU: 67ms
     CGroup: /system.slice/apache2.service
             ├─9156 /usr/sbin/apache2 -k start
             ├─9158 /usr/sbin/apache2 -k start
             └─9159 /usr/sbin/apache2 -k start

Feb 28 21:58:37 sabubhuju systemd[1]: Starting apache2.service - The Apache HTTP Server...
```

### b. Create Document Root Folders and set Permissions

```
sabubhuju@sabubhuju:~$ sudo mkdir -p /var/www/host1
sabubhuju@sabubhuju:~$ sudo mkdir -p /var/www/host2
sabubhuju@sabubhuju:~$ sudo chown -R $USER:$USER /var/www/host1
sabubhuju@sabubhuju:~$ sudo chown -R $USER:$USER /var/www/host2
sabubhuju@sabubhuju:~$ sudo chmod -R 755 /var/www
sabubhuju@sabubhuju:~$
```

### c. Create Sample HTML Files

```
sabubhuju@sabubhuju:~$ echo "<h1>Welcome Sabu Bhuju from Host1</h1>" | sudo tee /var/www/host1
/index.html
<h1>Welcome Sabu Bhuju from Host1</h1>
sabubhuju@sabubhuju:~$ echo "<h1>Welcome Sabu Bhuju from Host2</h1>" | sudo tee /var/www/host2
/index.html
<h1>Welcome Sabu Bhuju from Host2</h1>
sabubhuju@sabubhuju:~$
```

### d. Configure Virtual Hosts

```
sabubhuju@sabubhuju:~$ sudo nano /etc/apache2/sites-available/host1.example.com.conf
sabubhuju@sabubhuju:~$ sudo cat /etc/apache2/sites-available/host1.example.com.conf
<VirtualHost *:80>
    ServerAdmin admin@host1.example.com
    ServerName host1.example.com
    ServerAlias www.host1.example.com
    DocumentRoot /var/www/host1
    ErrorLog ${APACHE_LOG_DIR}/host1_error.log
    CustomLog ${APACHE_LOG_DIR}/host1_access.log combined
</VirtualHost>
sabubhuju@sabubhuju:~$ sudo nano /etc/apache2/sites-available/host2.example.com.conf
sabubhuju@sabubhuju:~$ sudo cat /etc/apache2/sites-available/host1.example.com.conf
<VirtualHost *:80>
    ServerAdmin admin@host1.example.com
    ServerName host1.example.com
    ServerAlias www.host1.example.com
    DocumentRoot /var/www/host1
    ErrorLog ${APACHE_LOG_DIR}/host1_error.log
    CustomLog ${APACHE_LOG_DIR}/host1_access.log combined
</VirtualHost>
sabubhuju@sabubhuju:~$
```

### e. Enable Virtual Hosts

```
sabubhuju@sabubhuju:~$ sudo a2ensite host1.example.com.conf
Site host1.example.com already enabled
sabubhuju@sabubhuju:~$ sudo a2ensite host2.example.com.conf
Enabling site host2.example.com.
To activate the new configuration, you need to run:
  systemctl reload apache2
sabubhuju@sabubhuju:~$ sudo systemctl reload apache2
```

### f. Update Hosts File for Testing

```
sabubhuju@sabubhuju:~$ sudo nano /etc/hosts
sabubhuju@sabubhuju:~$ sudo etc /etc/hosts
sudo: etc: command not found
sabubhuju@sabubhuju:~$ sudo cat /etc/hosts
127.0.0.1 localhost
127.0.1.1 sabubhuju

127.0.0.1 host1.example.com
127.0.0.1 host2.example.com

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

### g. Test Configuration

```
sabubhuju@sabubhuju:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using
 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Syntax OK
sabubhuju@sabubhuju:~$
```

### h. Test the Web Server

- Open a browser on the client machine:

http://host1.example.com

http://host2.example.com

## 9. Configure a Mail Server to receive and deliver the mail for the domain "example.com". The mail server should reject any email which has the content "SPAM" in the body of the email.

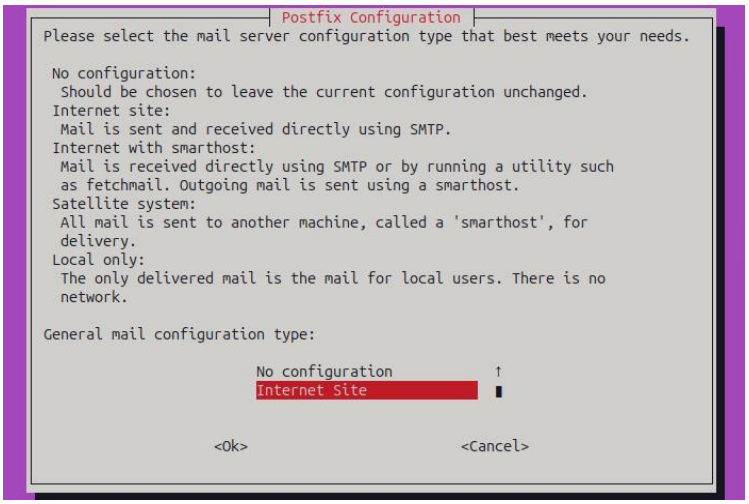
To configure a mail server to send and receive emails for the domain example.com, and reject emails containing the word "SPAM" in the body.

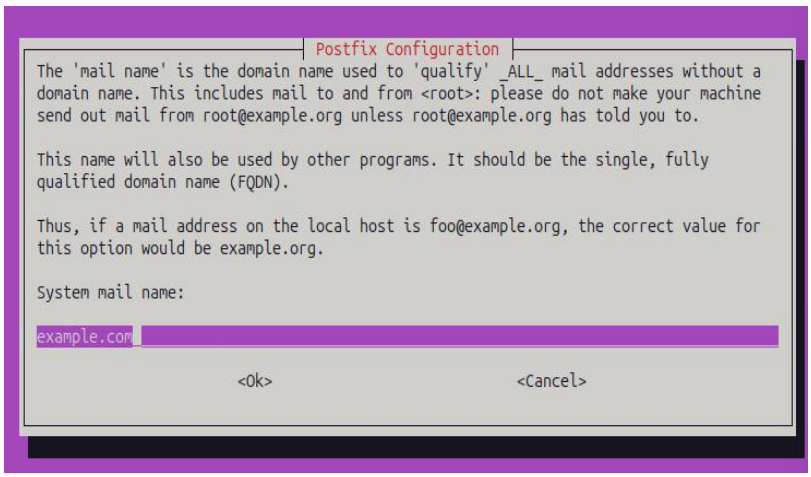
### a. Install Postfix

```
sabubhuju@sabubhuju:~$ sudo apt install postfix -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libnsl2
Suggested packages:
  mail-reader postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver
  postfix-mysql postfix-pcre postfix-pgsql postfix-sqlite procmail sasl2-bin
  | dovecot-common
The following NEW packages will be installed:
  libnsl2 postfix
0 upgraded, 2 newly installed, 0 to remove and 13 not upgraded.
```

During installation:

- Select **Internet Site**
- System mail name: example.com



**b. Configure Postfix Main Settings**

```
sabubhuju@sabubhuju:~$ sudo nano /etc/postfix/main.cf
sabubhuju@sabubhuju:~$ sudo cat /etc/postfix/main.cf
myhostname = mail.example.com
mydomain = example.com
myorigin = /etc/mailname
mydestination = $myhostname, localhost.$mydomain, localhost, sabubhuju, $mydomain
inet_interfaces = all
inet_protocols = all
home_mailbox = Maildir/
```

**c. Create body check rule:**

```
sabubhuju@sabubhuju:~$ sudo nano /etc/postfix/body_checks
sabubhuju@sabubhuju:~$ sudo cat /etc/postfix/body_checks
/SPAM/ REJECT Email contains forbidden content
```

**Enable body checks:**
**Edit etc/postfix/main.cf:**
**body_checks = regexp:/etc/postfix/body_checks**

```
body_checks = regexp:/etc/postfix/body_checks
sabubhuju@sabubhuju:~$
```

**d. Restart Postfix**

```
sabubhuju@sabubhuju:~$ sudo systemctl restart postfix
sabubhuju@sabubhuju:~$ sudo systemctl enable postfix
Synchronizing state of postfix.service with SysV service script with /usr
/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable postfix
sabubhuju@sabubhuju:~$
```

**e. Test the Mail Server**

```
sabubhuju@sabubhuju:~$ echo "Hello, this is a test mail" | mail -s "Test" user@example.com
sabubhuju@sabubhuju:~$ echo "This message contains SPAM" | mail -s "Spam test" user@example.com
sabubhuju@sabubhuju:~$ ls ~/Maildir/new
1772298326.V802I12078cM369057.sabubhuju
```

## f. Check logs

```
sabubhuju@sabubhuju:~$ echo "Hello, this is a test mail" | mail -s "Test" user@example.com
sabubhuju@sabubhuju:~$ echo "This message contains SPAM" | mail -s "Spam test" user@example.com
sabubhuju@sabubhuju:~$ ls ~/Maildir/new
1772298326.V802I12078cM369057.sabubhuju
sabubhuju@sabubhuju:~$ sudo tail -f /var/log/mail.log
2026-02-28T22:49:41.824738+05:45 sabubhuju postfix/local[13568]: BEEBE1631FC: to=<user@example.co
m>, relay=local, delay=0.06, delays=0.03/0.01/0/0.02, dsn=4.0.0, status=deferred (user lookup err
or)
2026-02-28T22:50:26.344454+05:45 sabubhuju postfix/pickup[13243]: 53F8F163294: uid=1000 from=<sab
ubhuju@sabubhuju>
2026-02-28T22:50:26.344780+05:45 sabubhuju postfix/cleanup[13566]: 53F8F163294: message-id=<20260
228170526.53F8F163294@mail.example.com>
2026-02-28T22:50:26.344953+05:45 sabubhuju postfix/cleanup[13566]: 53F8F163294: reject: body This
 message contains SPAM from local; from=<sabubhuju@sabubhuju>: 5.7.1 Email contains forbidden con
tent
2026-02-28T22:50:26.356369+05:45 sabubhuju postfix/cleanup[13566]: 53F8F163294: to=<user@example.
com>, relay=none, delay=0.02, delays=0.02/0/0/0, dsn=5.7.1, status=bounced (Email contains forbid
den content)
2026-02-28T22:50:26.365238+05:45 sabubhuju postfix/cleanup[13592]: 58C8E163297: message-id=<20260
228170526.58C8E163297@mail.example.com>
2026-02-28T22:50:26.368760+05:45 sabubhuju postfix/bounce[13591]: 53F8F163294: sender non-deliver
y notification: 58C8E163297
2026-02-28T22:50:26.369157+05:45 sabubhuju postfix/qmgr[13244]: 58C8E163297: from=<>, size=2238,
nrcpt=1 (queue active)
2026-02-28T22:50:26.374315+05:45 sabubhuju postfix/local[13568]: 58C8E163297: to=<sabubhuju@sabub
huju>, relay=local, delay=0.02, delays=0.01/0/0/0.01, dsn=2.0.0, status=sent (delivered to maildi
r)
2026-02-28T22:50:26.374467+05:45 sabubhuju postfix/qmgr[13244]: 58C8E163297: removed
```